

Alisa Hamilton¹, Hongru Du², Cliff Kerr³, Lauren Gardner¹

¹Johns Hopkins University, ²University of Virginia, ³Institute for Disease Modeling

Objective

We incorporate an adaptive decision framework into an agent-based model (ABM) made with Starsim¹ and run simulations on a Census-based synthetic population made using the open-source software GeoPops². During epidemics, individuals adjust their behavior based on income, health risk, and other personal factors, leading to variations in health and economic outcomes across demographic and geographic subgroups. Models that do not capture subgroup variation may overestimate intervention effectiveness, leading to inefficient resource use and blanket policies that can exacerbate disparities. In this tutorial, we demonstrate an example of a generic upper respiratory disease epidemic in which individuals weigh the health-wealth trade-offs of staying home from work. We apply this framework to a population of Howard County, MD and disaggregate outcomes by age group and Census Block Group (CBG).

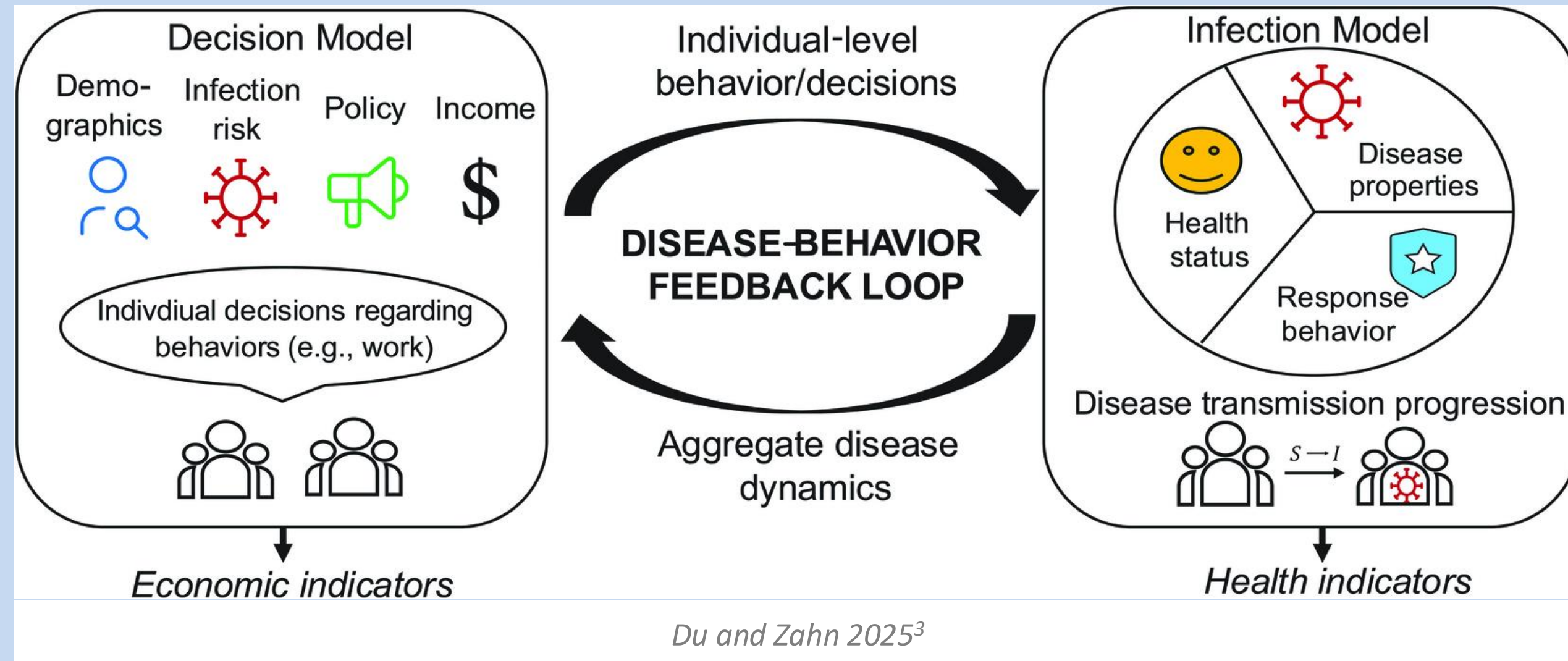


GitHub
Tutorial

Alisa Hamilton, MA, MS
PhD Candidate, Systems Engineering
Johns Hopkins University
ahamil33@jh.edu

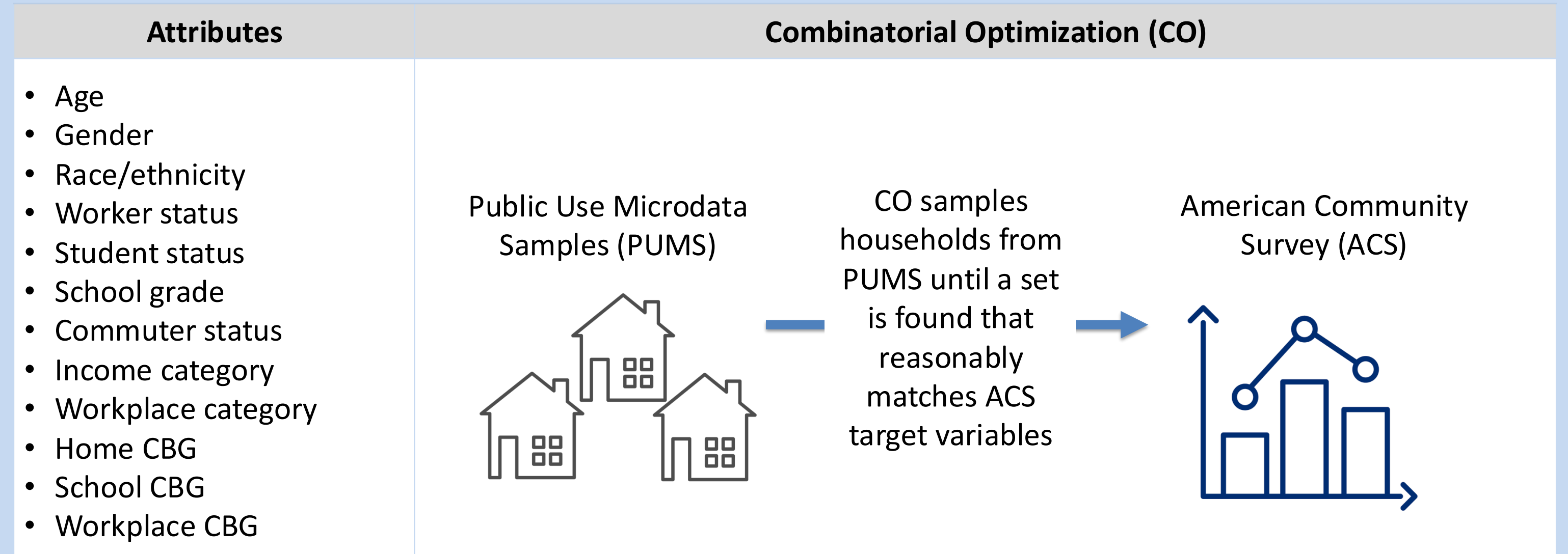
Dynamic Decision Making

Conceptual Framework

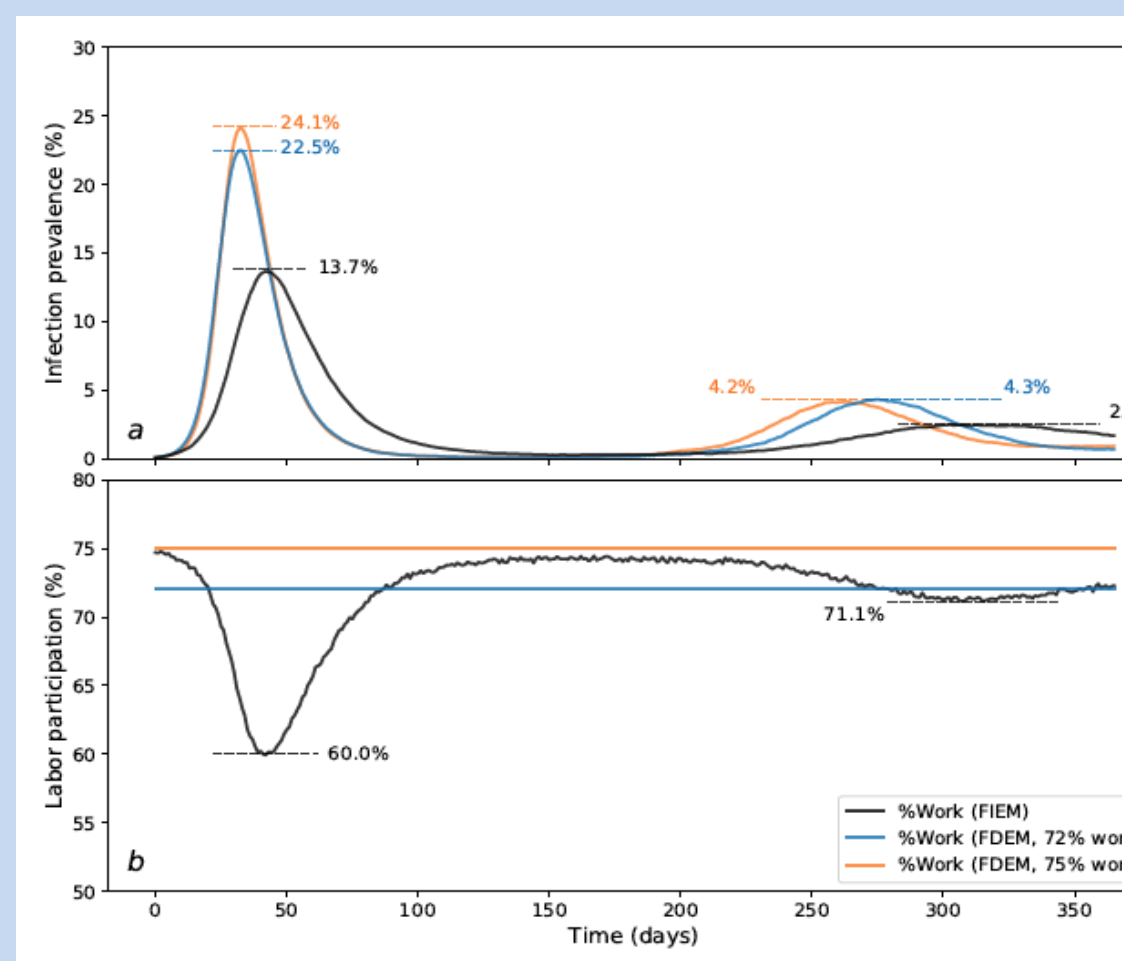


Synthetic Population with GeoPops

Step 1: Generate individuals and households



Compartmental Version



Feedback-Informed Epidemiological Model³
8x8 contact matrix
SES (low, high), disease vulnerability (low, high),
working status (yes, no)

Utility Framework

1. Calculate probability $S \rightarrow I$ for each risk group
2. Calculate utility values of working/not working
3. Bellman equation returns probabilities of working/not working
4. Update working status

Step 2: School and Workplace Assignment

	Methods	Data	Data Sources
Schools	- Students assigned to closest school that offers required grade - Teachers assigned to school by selecting teachers from nearby CBGs	Student and staff counts for each grade level in public schools	National Center for Education Statistics (NCES)
Workplaces	- Iterative proportional fitting (IPF) to make origin-destination commute matrix - IPF to make industry by destination commute matrix for each origin CBG - Includes outside workers	Origin-destination and workplace category matrices	Longitudinal Employer-Household Dynamics (LEHD)

ABM Adaptation

Components passed into Starsim to model disease transmission

SIRS model
Infection duration ~7 days
Transmission coefficient ~0.6
Initial prevalence ~0.001
Waning ~0.008

Feedback module
Removes edges from work network when agents decide to stay home from work

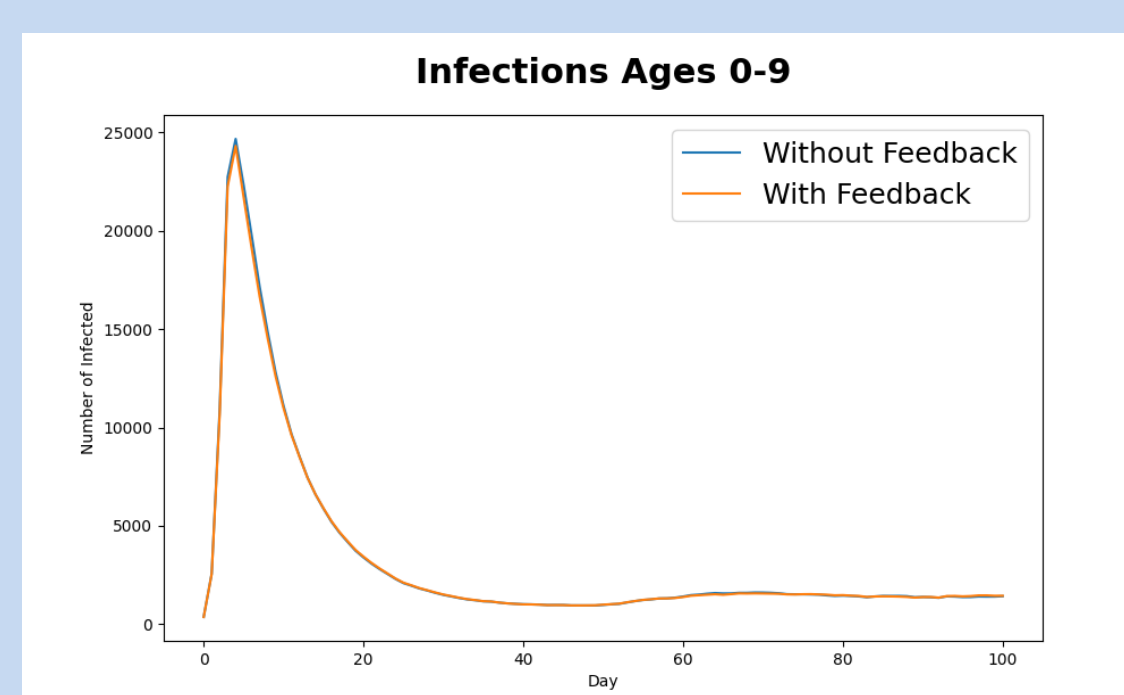
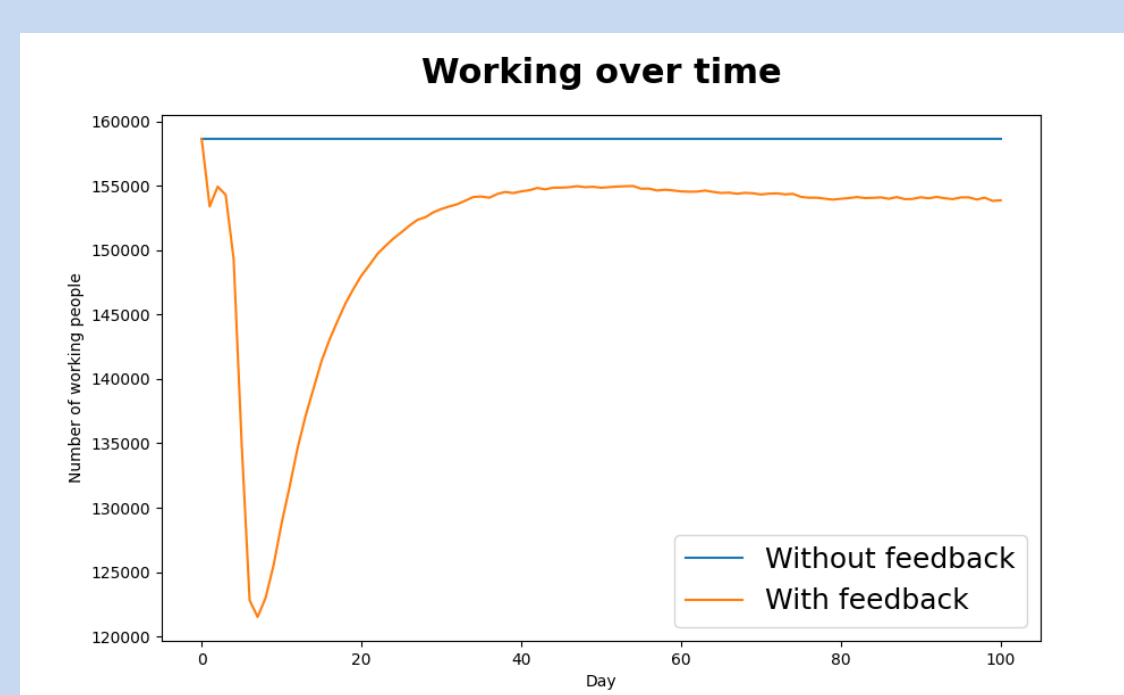
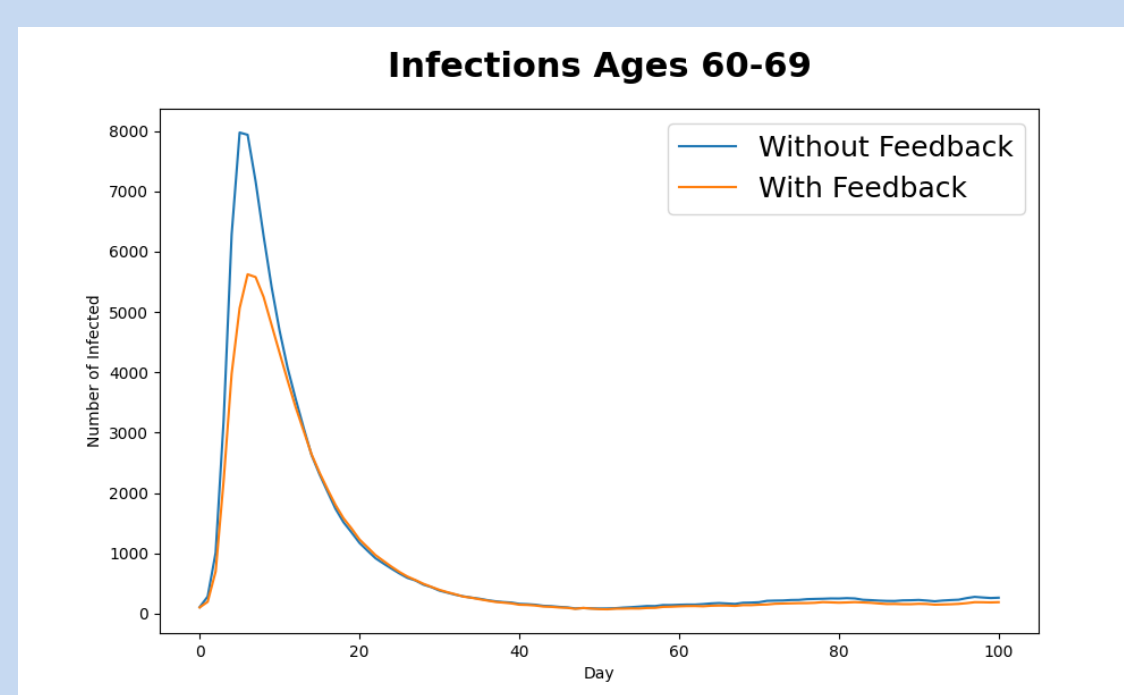
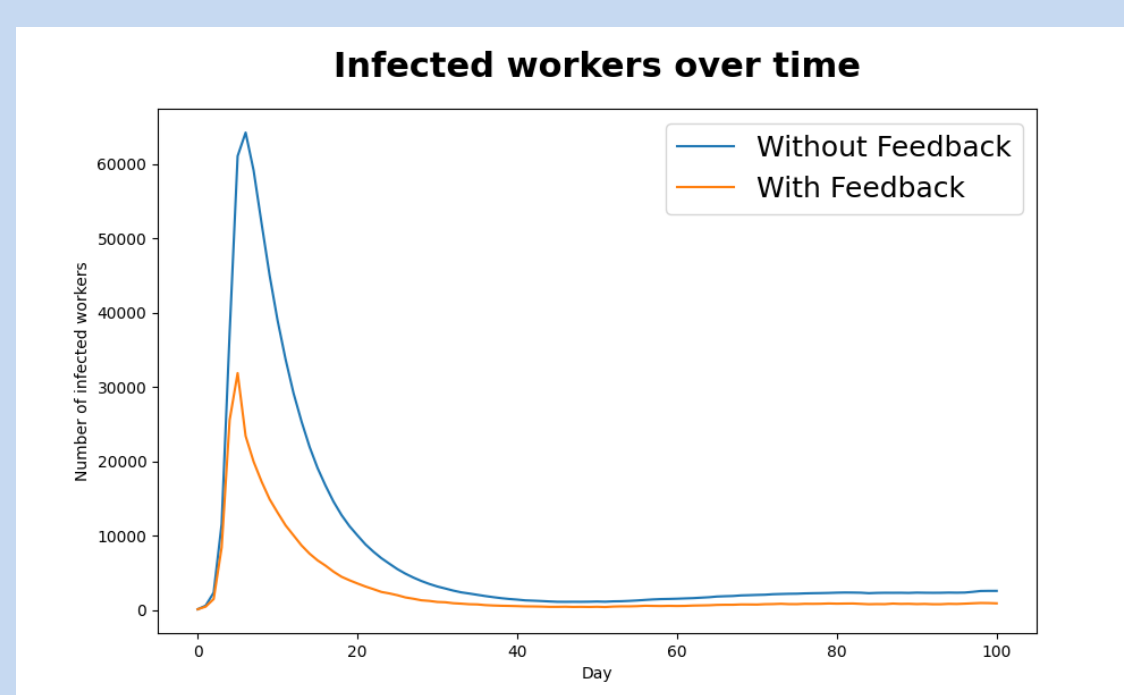
GeoPops
SES: Income (>40k, <=40k)
Vulnerability: Age (>65, <=65)

Subgroup Tracking
Infections by age and CBG

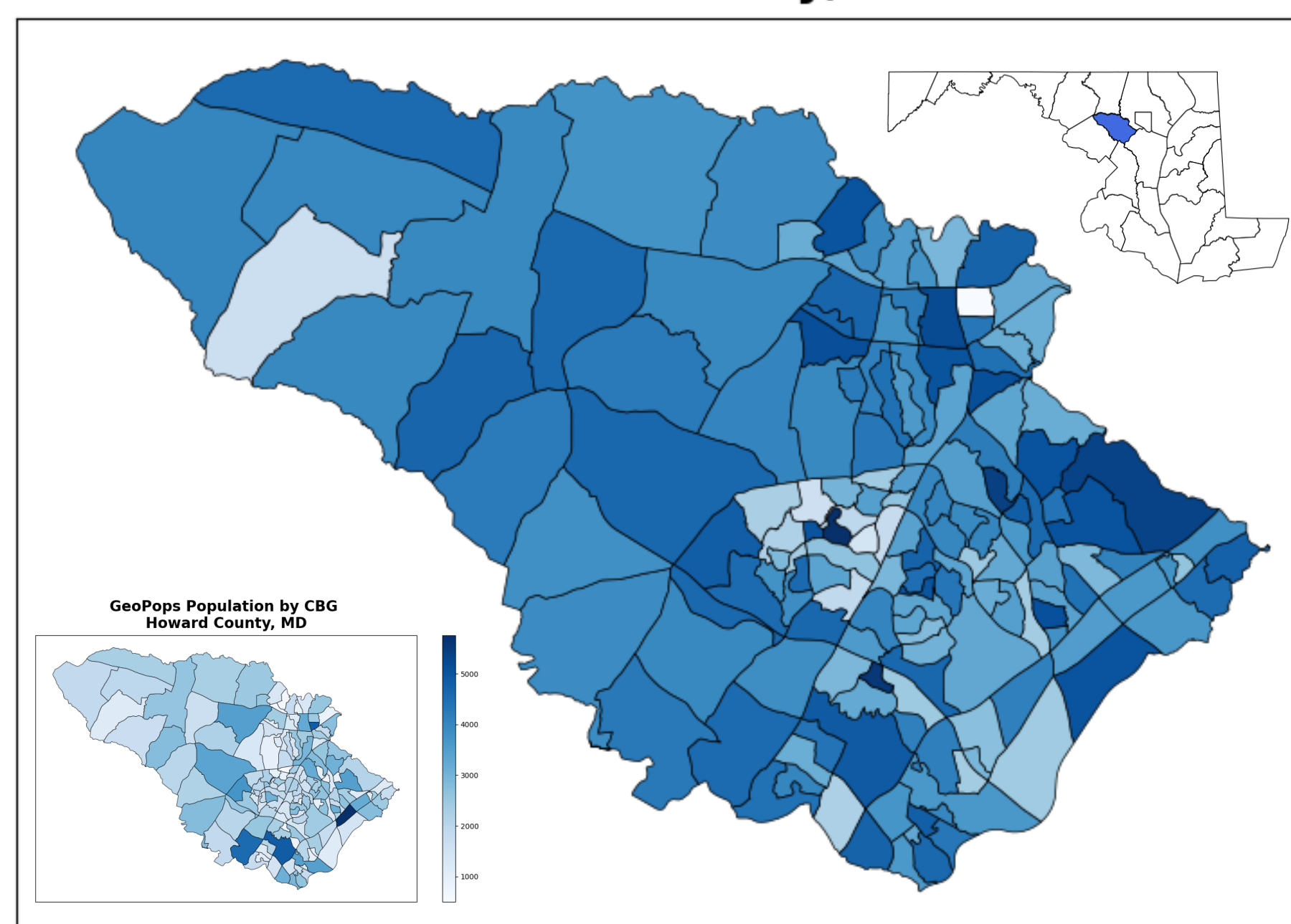
Step 3: Network Generation

Network	Algorithm	Parameters
Household	All agents within a household are fully connected	N/A
School	Stochastic Block Model within each school	- Assortativity: $\alpha=0.9$ - Mean degree: $K=12$ - Blocks are grade levels (Pre-k – 12)
Workplace	Stochastic Block Model within each workplace	- Assortativity: $\alpha=0.9$ - Mean degree: $K=8$ - Blocks are income levels (<40k or $\geq 40k$)
Group quarters	Watts-Strogatz Model within each GQ facility	- Mean degree: $K=12$ - Rewiring probability: $\beta=0.25$

Example with Howard County, MD*



Total Infections per 1000 Population by CBG Howard County, MD



* This example reflects a simple application with a generic upper respiratory infection in Howard County, MD (N ~340,000)

Next Steps

- Incorporate work from home
- More detailed COVID-19 model: SEIRD with age-dependent infectiousness and susceptibility
- Run on population of Maryland
- Compare to observed data for first wave of the pandemic
- Test policies (e.g., paid sick leave, cash transfer)

References

1. Hamilton A, Tulchinsky A, Lin G, Kerr C, Klein E, and Gardner L. 2025. GeoPops: Geographically Realistic Synthetic Population Generator. Johns Hopkins University and One Health Trust. <https://github.com/ACCIDDA/GeoPops>.
2. Kerr C, Stuart R, A Romesh, et al. 2024. "Starsim: A Fast, Flexible Toolbox for Agent-Based Modeling of Health and Disease." <https://starsim.org/>.
3. Du H, Zahn MV, Loo SL, Alleman TW, Truelove S, et al. (2025) Improving policy design and epidemic response using integrated models of economic choice and disease dynamics with behavioral feedback. PLOS Computational Biology 21(10): e1013549. <https://doi.org/10.1371/journal.pcbi.1013549>

github.com/ACCIDDA/GeoPops/tutorials/Epidemics

Alisa Hamilton, MA, MS
PhD Candidate, Systems Engineering
Johns Hopkins University
ahamil33@jh.edu